Digital Forensics Investigation Report

1. Introduction

This project focuses on the forensic analysis of a suspicious email log file to identify unauthorized access and possible malicious activity in a system. The investigation aims to detect evidence of intrusion, analyze system behavior, and reconstruct the sequence of events.

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2. Objective

- To analyze email log files
- To identify suspicious activities
- To detect unauthorized system access
- To create a timeline of events
- To prepare a forensic report

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3. Tools Used

- Ubuntu Terminal
- Linux Commands (`cat`, `grep`, `ls`)

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4. Evidence Collection

The evidence used in this investigation is an email log file:

File Name: logfile-2012-07-09-1300.eml

This file contains system-generated logs including user activity, timestamps, and command execution records.

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5. Analysis & Findings

5.1 Email Analysis

- The system sent logs to: `joe.sum.twelve@gmail.com`
- The sender was: `root@tracys-macbook-air.local`

This indicates that a system administrator (root user) was involved, which is highly sensitive.

5.2 Login Activity

- User `tracysumtwelve` logged into the system multiple times:
  - 09:22 AM
  - 10:42 AM

Repeated login activity indicates possible suspicious access.

5.3 Command Execution

The following commands were found in the log:

- `cd Documents`
- `ls`
- `zip -e documents.zip`

This suggests:

- Navigation through files
- Listing directory contents
- Creating a password-protected ZIP file
- This may indicate **data collection and compression for exfiltration**

5.4 Suspicious Data

- Strange text patterns and email strings detected:
  - `gmail.ccoralbluetwo@hotmail.com`
  - `legalBee`

These may represent:

- Hidden credentials
- Encoded or manipulated data

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6. Timeline of Events

Time	Event
09:04 AM	System started
09:22 AM	User login detected
After login	Commands executed (cd, ls, zip)
10:41 AM	System restarted
10:42 AM	User logged in again

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7. Result & Conclusion

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ritesh-jha@ritesh-jha-Inspiron-15-3520:~/Desktop/email$ grep "@" logfile-2012-07-09-1300.eml
Delivered-To: joe.sum.twelve@gmail.com
Return-Path: <root@tracys-macbook-air.local>
Received-SPF: neutral (google.com: 2600:1003:b023:1c90:1907:36f8:5fa2:8f64 is neither permitted nor denied by best guess record for domain of root@tracys-macbook-air.local) client-ip=2
600:1003:b023:1c90:1907:36f8:5fa2:8f64;
Authentication-Results: mx.google.com; spf=neutral (google.com: 2600:1003:b023:1c90:1907:36f8:5fa2:8f64 is neither permitted nor denied by best guess record for domain of root@tracys-m
acbook-air.local) smtp.mail=root@tracys-macbook-air.local
To: joe.sum.twelve@gmail.com
Message-Id: <20120709170000.2FEB3693F3@Tracys-MacBook-Air.local>
From: root@Tracys-MacBook-Air.local (System Administrator)
gmail.ccoralbluetwo@hotmail.com<tab><tab>thingas<del><del>s<tab>somethingss<del>legalBee
ritesh-jha@ritesh-jha-Inspiron-15-3520:~/Desktop/email$ grep -i "login" logfile-2012-07-09-1300.eml
ritesh-jha@ritesh-jha-Inspiron-15-3520:~/Desktop/email$ grep -i "Mon" logfile-2012-07-09-1300.eml
        Mon, 9 Jul 2012 10:00:04 -0700 (PDT)
        Mon, 09 Jul 2012 10:00:03 -0700 (PDT)
        Mon, 09 Jul 2012 10:00:03 -0700 (PDT)
        id 2FEB3693F3; Mon,  9 Jul 2012 13:00:00 -0400 (EDT)
Date: Mon,  9 Jul 2012 13:00:00 -0400 (EDT)
![LogKext Daemon created new logfile : Fri Jul  6 14:01:54 2012]
![LogKext Daemon starting up : Mon Jul  9 09:04:09 2012]
![User 'tracysumtwelve' has logged in : Mon Jul  9 09:22:10 2012]
![LogKext Daemon starting up : Mon Jul  9 10:41:45 2012]
![User 'tracysumtwelve' has logged in : Mon Jul  9 10:42:27 2012]
ritesh-jha@ritesh-jha-Inspiron-15-3520:~/Desktop/email$
```

The forensic analysis reveals that:

- Unauthorized access may have occurred
- Root-level involvement was detected
- Multiple login sessions were observed
- File access and compression activity took place

These findings strongly indicate a **possible data exfiltration attempt**

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8. Recommendations

- Restrict root user access
- Monitor login activity regularly
- Use strong authentication mechanisms
- Implement intrusion detection systems
- Perform regular security audits

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9. Final Remark

This investigation demonstrates how forensic techniques can be used to detect cyber incidents and analyze suspicious system behavior effectively.